C++ Functions Lab Practice

void Functions (No Parameters, No Return)

Note: This lab uses a completely different problem set to reinforce the same concept — defining and calling void functions with no parameters and no return value.

Objectives

- Write void functions that display real-world themed output.
- Call functions from main() in a meaningful sequence.
- Use loops and cout inside function bodies for patterns and tables.
- Combine multiple void functions to build complete mini-programs.
- Practice function prototypes when defining functions after main().

Part 1: Real-World Print Functions

Note: Questions 1 and 2 include reference code. Write your own code for Questions 3–8.

EASY — Single output functions

Q1. Define printSchoolName() that displays a school name banner. Call it from main(). (Code provided as reference.)

Expected Output:

```
** Bright Future Academy **
```

C++ Code:

```
void printSchoolName() {
    cout << "** Bright Future Academy **" << endl;
}

int main() {
    printSchoolName();
    return 0;
}
```

Q2. Define printATMWelcome() that prints an ATM welcome screen. Call it from main(). (Code provided as reference.)

Expected Output:

```
################################
#   Welcome to QuickBank ATM   #
#   Please insert your card.   #
```

```
################################
```

C++ Code:

```
void printATMWelcome() {
    cout << "################################" << endl;
    cout << "#   Welcome to QuickBank ATM   #" << endl;
    cout << "#   Please insert your card.   #" << endl;
    cout << "################################" << endl;
}

int main() {
    printATMWelcome();
    return 0;
}
```

Q3. Define printDayOfWeek() that prints the 7 days of the week, one per line. Call it from main().

Expected Output:

```
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
```

Q4. Define printCountdown() that counts down from 5 to 1 then prints 'Go!'. Call it from main().

Expected Output:

```
5
4
3
2
1
Go!
```

Q5. Define printReceipt() that prints a simple store receipt (3 fixed items with prices and a total). Call it from main().

Expected Output:

```
------- RECEIPT -------
Milk        Rs. 120
```

```
Bread       Rs. 80
Eggs        Rs. 200
-----------------------
Total:      Rs. 400
```

MEDIUM — Loops and patterns inside functions

Q6. Define printEvenNumbers() that prints all even numbers from 2 to 20. Call it from main().

Expected Output:

```
2  4  6  8  10  12  14  16  18  20
```

Q7. Define printCheckerRow() that prints one row of alternating # and - characters (10 wide). Call it 4 times from main() to build a checkerboard pattern.

Expected Output:

```
#-#-#-#-#-
#-#-#-#-#-
#-#-#-#-#-
#-#-#-#-#-
```

Q8. Define printTemperatureTable() that prints a conversion table from Celsius to Fahrenheit for 0, 10, 20, 30, 40, and 50 degrees Celsius. Use the formula F = C * 9/5 + 32 with fixed values hard-coded. Call it from main().

Expected Output:

```
Celsius   Fahrenheit
--------------------
0         32
10        50
20        68
30        86
40        104
50        122
```

---

void Functions — Key Rules

Situation	Syntax / Rule

define a void function	`void funcName() { ... }`
call a void function	`funcName();`
print output inside function	`cout << value << endl;`
use a loop inside a function	`for(int i=...) { cout << ...; }`
build patterns with loops	`nested for loops inside function body`
call same function N times	`write funcName(); N times in main()`

Part 2: Function Sequences and Prototypes

Note: Questions 1 and 2 include reference code. Write your own code for Questions 3–9.

EASY — Sequencing and reuse

Q1. Define showOpeningMessage() and showClosingMessage(). Call them at the start and end of main(), with some action in between. (Code provided as reference.)

Expected Output:

```
--- Program Started ---
Processing data...
--- Program Ended ---
```

C++ Code:

```
void showOpeningMessage() {
    cout << "--- Program Started ---" << endl;
}
void showClosingMessage() {
    cout << "--- Program Ended ---" << endl;
}

int main() {
    showOpeningMessage();
    cout << "Processing data..." << endl;
    showClosingMessage();
    return 0;
}
```

Q2. Define printCopyright() after main() using a prototype. Call it at the end of main(). (Code provided as reference.)

Expected Output:

```
(c) 2026 My C++ Program. All rights reserved.
```

C++ Code:

```
void printCopyright();   // prototype

int main() {
    printCopyright();
    return 0;
}

void printCopyright() {
    cout << "(c) 2026 My C++ Program. All rights reserved." << endl;
}
```

Q3. Define printMorningGreeting() and printEveningGreeting(). Call printMorningGreeting() twice and printEveningGreeting() once from main().

Expected Output:

```
Good Morning! Have a great day.
Good Morning! Have a great day.
Good Evening! Time to rest.
```

Q4. Define printPageBreak() that prints a blank line followed by a row of 40 equals signs. Call it between three section labels printed directly in main().

Expected Output:

```
Section 1

========================================
Section 2

========================================
Section 3
```

MEDIUM — Multi-function mini programs

Q5. Define printInvoiceHeader(), printInvoiceItems(), and printInvoiceTotal(). Call all three in order from main() to produce a complete invoice.

Expected Output:

```
======= INVOICE =======
Item          Qty   Price
-----------------------
Pen            2    Rs.40
```

```
Notebook       1    Rs.150
USB Cable      3    Rs.360
-----------------------
TOTAL:              Rs.550
```

Q6. Define printNumberSquares() that prints the squares of numbers 1 through 8 in a table. Call it from main().

Expected Output:

```
Number   Square
---------------
1        1
2        4
3        9
4        16
5        25
6        36
7        49
8        64
```

Q7. Define printDiamondTop() and printDiamondBottom() that together print a diamond shape made of asterisks. Call both from main().

Expected Output:

```
  *
 ***
*****
 ***
  *
```

HARD — Full programs using only void functions

Q8. Define void functions to read a phone book file and display each entry formatted. Use: openFile(), printPhoneBookHeader(), readAndDisplayEntries(), printPhoneBookFooter(), and closeFile(). Define printPhoneBookFooter() after main() with a prototype.

File Contents (phonebook.txt):

```
AliHassan 0300-1234567
SaraMalik 0311-9876543
UsmanKhan 0321-4567890
…. number of records are unknown
```

Expected Screen Output:

```
=========== PHONE BOOK ===========
Name: AliHassan     | 0300-1234567
Name: SaraMalik     | 0311-9876543
Name: UsmanKhan     | 0321-4567890
==================================

Total Entries: 3
```

Q9. Write a complete program using only void functions. Write a weekly class schedule to schedule.txt, then read it back and display it formatted on screen.

Expected File Contents (schedule.txt):

```
Monday C++Programming 9:00AM
Wednesday DataStructures 11:00AM

Friday Mathematics 2:00PM
```

Expected Screen Output (after reading back):

```
======= WEEKLY CLASS SCHEDULE =======
Monday    : C++ Programming   9:00 AM
Wednesday : Data Structures  11:00 AM
Friday    : Mathematics       2:00 PM

=====================================
```

Function Prototypes — Key Rules

Situation	Syntax / Rule
function before main	`no prototype needed`
function after main	`void funcName();  above main`
prototype has semicolon	`void funcName();  // NO body`
calling order in main()	`call functions in any logical order`
reusing a function	`call funcName(); as many times as needed`
output stays inside function	`never cout outside the function for its job`

Excellent Work!